Numpy---it is a modeule numpy module has 217 function module divide 2 part inbuild module user defined module matrix--library which handle matrix--array(numpy module handle this) image--array| array-image text--array| array--text numpy library we will be using in---data analysis, image processing, uses of numpy 1. arithmatic operations 2. statistical operator 3. bitwise opertor 4. copy array, view array 5. stacking 6. matrix operation 7. linear algebra 8. broadcating 9. math operations 10. searching, sorting and counting 1d array vs 2d array vs 3d array(multidimentional) Numpy-- library which handle multidimentional array 1d--1 rows 2d-- 2 rows 3d or nd--- multiple coulmns and rows

In [1]:
```
import numpy as np
```

In [2]:
```
np.__version__
```

Out[2]:
```
'1.26.4'
```

In [3]:
```
my_list = [0,1,2,3,4,5]
my_list
```

Out[3]:
```
[0, 1, 2, 3, 4, 5]
```

In [4]:
```
type(my_list)
```

Out[4]:
```
list
```

In [5]:
```
arr = np.array(my_list)
arr
```

Out[5]:
```
array([0, 1, 2, 3, 4, 5])
```

In [6]:
```
type(arr)
```

Out[6]:
```
numpy.ndarray
```

In [8]:
```
np.arange(10)  # it will print the value within interval
```

Out[8]:
```
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

In [9]:
```
np.arange(10,15)
```

Out[9]:
```
array([10, 11, 12, 13, 14])
```

In [10]:
```
np.arange(10,20,5)
```

Out[10]:
```
array([10, 15])
```

In [11]:
```
np.arange(10,20,5,2)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[11], line 1
----> 1 np.arange(10,20,5,2)

TypeError: Cannot interpret '2' as a data type
```

In [14]:
```
np.zeros(5) # parameter tuning
```

Out[14]:
```
array([0., 0., 0., 0., 0.])
```

In [15]:
```
np.zeros(5, dtype=int)   # hyperparameter tuning---change system parameter
```

Out[15]:
```
array([0, 0, 0, 0, 0])
```

In [16]:
```
np.zeros([2,2], dtype=int)  # 2d array
```

Out[16]:
```
array([[0, 0],
       [0, 0]])
```

In [18]:
```
np.zeros([5,10], dtype=int)  # multidimentional array # np.zeror()=== print only
```

Out[18]:
```
array([[0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0]])
```

In [20]:
```
np.ones(3)
```

Out[20]:
```
array([1., 1., 1.])
```

In [21]:
```
np.ones([3,5], dtype=int)
```

Out[21]:
```
array([[1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1]])
```

In [22]:
```
np.ones([3,5])
```

Out[22]:
```
array([[1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.]])
```

In [23]:
```
np.arange(20,10) # because 1st argument is grater than 2nd
```

Out[23]:
```
array([], dtype=int32)
```

In [24]:
```
np.arange(10, 20)  # np.arrange---1st argument< 2nd argument
```

Out[24]:
```
array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])
```

In [25]:
```
np.arange(-20, 10)
```

Out[25]:
```
array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11, -10,  -9,  -8,
        -7,  -6,  -5,  -4,  -3,  -2,  -1,   0,   1,   2,   3,   4,   5,
         6,   7,   8,   9])
```

In [26]:
```
np.twos(3) # twos--- is not function of numpy
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[26], line 1
----> 1 np.twos(3)

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:333, in __getattr__(attr)
    330     "Removed in NumPy 1.25.0"
    331     raise RuntimeError("Tester was removed in NumPy 1.25.")
--> 333 raise AttributeError("module {!r} has no attribute "
    334                      "{!r}".format(__name__, attr))

AttributeError: module 'numpy' has no attribute 'twos'
```

In [27]:
```
np.threes(3) # AttributeError
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[27], line 1
----> 1 np.threes(3)

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:333, in __getattr__(attr)
    330     "Removed in NumPy 1.25.0"
    331     raise RuntimeError("Tester was removed in NumPy 1.25.")
--> 333 raise AttributeError("module {!r} has no attribute "
    334                      "{!r}".format(__name__, attr))

AttributeError: module 'numpy' has no attribute 'threes'
```

In [28]:
```
random.rand(3,2)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[28], line 1
----> 1 random.rand(3,2)

NameError: name 'random' is not defined
```

In [29]:
```
np.random.rand(2) # it generate random numbers of float array
```

Out[29]:
```
array([0.41722191, 0.21198383])
```

In [30]:
```
np.random.rand(2)
```

Out[30]:
```
array([0.47303685, 0.49884501])
```

In [31]:
```
np.random.rand(5)
```

Out[31]:
```
array([0.73890137, 0.35346022, 0.19406871, 0.41552669, 0.56088774])
```

In [32]:
```
np.random.rand(3,2)
```

Out[32]:
```
array([[0.21931953, 0.34531496],
       [0.04004896, 0.45000597],
       [0.3692725 , 0.10941622]])
```

In [41]:
```
np.random.randint(3) # it will print random value between 0-2
```

Out[41]:
```
0
```

In [38]:
```
np.random.randint(5,10)
```

Out[38]:
```
6
```

In [43]:
```
np.random.randint(5,10,3)
```

Out[43]:
```
array([5, 8, 6])
```

In [45]:
```
ran =np.random.randint(5,10,3)
print(ran)
```

```
[9 9 8]
```

In [47]:
```
otp =np.random.randint(5,10,4) # np.random.randint----it generate random number
print(otp)
```

```
[9 6 7 6]
```

In [48]:
```
array1 = np.random.randint(10,40,(10,10))
print(array1)
```

```
[[25 28 23 35 22 21 39 30 28 31]
 [33 18 32 28 13 13 34 27 10 39]
 [36 10 13 25 39 25 19 25 30 23]
 [12 34 10 39 29 32 12 14 37 36]
 [33 16 13 35 24 39 33 29 32 19]
 [27 12 12 11 24 26 17 38 32 32]
 [26 18 39 25 17 36 16 38 27 29]
 [34 29 12 14 33 14 15 36 11 28]
 [38 36 16 17 32 34 20 39 21 36]
 [12 20 22 27 33 39 31 31 30 10]]
```

In [49]:
```
len(array1)  # gives 10 rows
```

Out[49]:
```
10
```

In [50]:
```
array1 = np.random.randint(10,40,(10,5))
print(array1)
```

```
[[11 27 37 19 11]
 [29 19 24 18 11]
 [34 34 15 16 26]
 [34 20 18 35 24]
 [25 26 34 30 10]
 [25 20 25 26 14]
 [14 36 39 24 25]
 [24 23 11 28 10]
 [13 28 38 11 29]
 [19 36 17 27 17]]
```

In [51]:
```
array1 = np.random.randint(10,40,(10,10))
print(array1)
```

```
[[39 30 28 35 33 30 13 15 10 15]
 [33 21 13 24 18 29 27 39 31 24]
 [11 16 27 14 28 38 12 38 16 24]
 [12 38 18 23 22 19 21 37 13 12]
 [12 21 26 25 27 21 26 15 11 23]
 [39 14 28 39 23 13 16 34 25 19]
 [14 32 21 38 19 32 10 25 36 21]
 [16 25 39 38 10 28 10 10 32 20]
 [10 36 24 35 22 21 17 28 37 16]
 [30 23 35 10 28 24 26 31 35 33]]
```

indexing and slicing in array

In [52]:
```
arr1 = np.random.randint(1,5,(3,3))
print(arr1)
```

```
[[1 4 1]
 [2 2 4]
 [3 3 2]]
```

In [53]:
```
len(arr1)
```

Out[53]:
```
3
```

In [54]:
```
arr1.shape
```

Out[54]:
```
(3, 3)
```

In [55]:
```
arr1[0]  # print row
```

Out[55]:
```
array([1, 4, 1])
```

In [57]:
```
arr1[:,1]
```

Out[57]:
```
array([4, 2, 3])
```

In [58]:
```
arr1[:,2]  # print columns
```

Out[58]:
```
array([1, 4, 2])
```

In [59]:
```
arr1
```

Out[59]:
```
array([[1, 4, 1],
       [2, 2, 4],
       [3, 3, 2]])
```

In [60]:
```
arr1[0]
```

Out[60]:
```
array([1, 4, 1])
```

In [61]:
```
arr1[0, 0]
```

Out[61]:
```
1
```

In [62]:
```
arr1[1, 2] # 1st row and 2nd column
```

Out[62]: 4

In [63]:
```
v = arr1[1, 2]
print(v)
```

```
4
```

In [65]:
```
arr1
```

Out[65]:
```
array([[1, 4, 1],
       [2, 2, 4],
       [3, 3, 2]])
```

In [64]:
```
arr1[-2]
```

Out[64]:
```
array([2, 2, 4])
```

In [66]:
```
arr1[1:2]
```

Out[66]:
```
array([[2, 2, 4]])
```

In [ ]: